

BUSTED at AraGenEval Shared Task: A Comparative Study of Transformer-Based Models for Arabic AI-Generated Text Detection

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Abstract

This paper details our submission to the AraGenEval Shared Task on Arabic AI-generated text detection, where our team, BUSTED, secured 5th place. We investigated the effectiveness of three pre-trained transformer models: AraELECTRA, CAMeLBERT, and XLM-RoBERTa. Our approach involved fine-tuning each model on the provided dataset for a binary classification task. Our findings revealed a surprising outcome: the multilingual XLM-RoBERTa model achieved the highest performance with an F1-score of 0.7701, outperforming the specialized Arabic models. This work underscores the complexities of AI-generated text detection and highlights the strong generalization capabilities of multilingual models.

1 Introduction

The increasing sophistication of large language models (LLMs) has blurred the line between human and machine-authored text. This reality poses significant societal risks, from accelerating the spread of misinformation to undermining academic integrity. In response, the development of reliable detectors for AI-generated text has become a pressing research priority. The AraGenEval Shared Task (Abudalfa et al., 2025) provides a crucial benchmark for this challenge in the Arabic language, a domain where such tools are still developing.

Our approach was to systematically evaluate the performance of different transformer architectures. We fine-tuned each model to perform binary classification, adapting their general linguistic knowledge to the specific task of distinguishing human from machine authorship. We specifically investigated:

1. **AraELECTRA** (Antoun et al., 2021), a specialized Arabic model.
2. **CAMeLBERT** (Inoue et al., 2021), a widely-used Arabic BERT model.

3. **XLM-RoBERTa** (Conneau et al., 2020), a large multilingual model.

This paper’s contributions are threefold. First, we provide a direct comparison of monolingual versus multilingual models for Arabic text detection. Second, we demonstrate that a multilingual model can achieve superior performance, a counter-intuitive but important finding. Finally, we analyze how certain preprocessing choices, such as aggressive text normalization, can inadvertently harm model performance by erasing subtle stylistic cues. Our best-performing model secured a 5th place finish in the shared task.

2 Related Work

Early efforts in authorship attribution and machine-text detection relied on statistical stylometry, using features like n-gram frequencies, readability scores, and syntactic structures to train classifiers. While effective for simpler models, these methods are less robust against the fluency of modern LLMs.

The current research landscape is dominated by neural network approaches. Fine-tuning pre-trained transformers like BERT (Devlin et al., 2019) has emerged as a powerful and accessible baseline. Other lines of inquiry focus on detecting statistical artifacts unique to the generative process of LLMs or embedding a "watermark" into the text during generation. Our work aligns with the fine-tuning paradigm and is inspired by comprehensive comparative studies like that of (Al-Shboul et al., 2024), applying a similar methodology to the specific and under-resourced domain of Arabic AI-text detection.

3 Background

3.1 Task Setup

The AraGenEval shared task is a binary text classification problem. The goal is to classify a given

Arabic text snippet as either ‘human-written’ or ‘machine-generated’.

- **Input:** A string of Arabic text.
- **Output:** A binary label (‘human’ or ‘machine’).

3.2 Dataset Analysis

The task utilized the AraGenEval dataset, which, after cleaning, contains 4,734 training samples. The class distribution is nearly balanced, with 2,399 samples (50.68%) labeled as ‘machine’ and 2,335 (49.32%) as ‘human’. Our initial analysis revealed several key distinguishing features within the training data:

Text Length: A significant discriminator is text length. Human-written texts are substantially longer on average (4059.13 characters) compared to machine-generated texts (1934.53 characters). This suggests that document length alone could be a strong, albeit potentially brittle, feature.

Lexical and N-gram Differences: We observed distinct topical and stylistic patterns.

- **Human-written texts** frequently contain words like *آغزة* (Gaza), *الحرب* (the war), and *إسرائيل* (Israel), and n-grams such as *الولايات المتحدة* (the United States), pointing to a focus on specific current geopolitical events.
- **Machine-generated texts** use more general and formal vocabulary, such as *أمكن* (can be), *أبشکل* (in a way), and n-grams like *المجتمع الدولي* (the international community) and *آحقوق الإنسان* (human rights), suggesting a more analytical or descriptive style.

These lexical and phraseological differences highlight the distinct registers and topics between the two classes, which are crucial for classification.

3.3 Related Work

Our work is built on the transformer architecture (Vaswani et al., 2017). Our comparative approach, which evaluates multiple deep learning models for an Arabic text classification task, is inspired by comprehensive surveys in the field, such as the

one conducted by (Al-Shboul et al., 2024). We specifically leverage pre-trained models including BERT (Devlin et al., 2019), ELECTRA (Clark et al., 2020), and XLM-RoBERTa (Conneau et al., 2020). Our chosen models, CAMELBERT (Inoue et al., 2021) and AraELECTRA (Antoun et al., 2021), are state-of-the-art for the Arabic language, while XLM-RoBERTa is a robust multilingual baseline.

4 System Overview

We implemented three systems based on different pre-trained models. Our overall workflow is illustrated in Figure 2.

4.1 System 1: AraELECTRA

This system uses ‘aubmindlab/araelectra-base-discriminator’. A key component was an aggressive Arabic text normalization preprocessing step applied before tokenization. This function normalized various Arabic characters (e.g., - آ، إ، أ، ؤ and - ة،) and stripped all Arabic diacritics and non-alphanumeric characters.

4.2 System 2: CAMELBERT

This system is based on ‘CAMEL-Lab/bert-base-arabic-camelbert-mix’. In contrast to the AraELECTRA system, we did not apply any specific text normalization, relying entirely on the model’s pre-trained tokenizer.

4.3 System 3: XLM-RoBERTa

Our third and best-performing system utilizes the multilingual ‘xlm-roberta-base’ model. Similar to the CAMELBERT setup, no language-specific normalization was performed.

5 Experimental Setup

5.1 Data Splits

The experimental setups for data splitting differed:

- **AraELECTRA & CAMELBERT:** We used the entire training dataset of 4,734 samples for both training and evaluation during the development phase.
- **XLM-RoBERTa:** We split the main training data into an 80% training set (3,787 samples) and a 20% validation set (947 samples), stratified to maintain the label distribution.

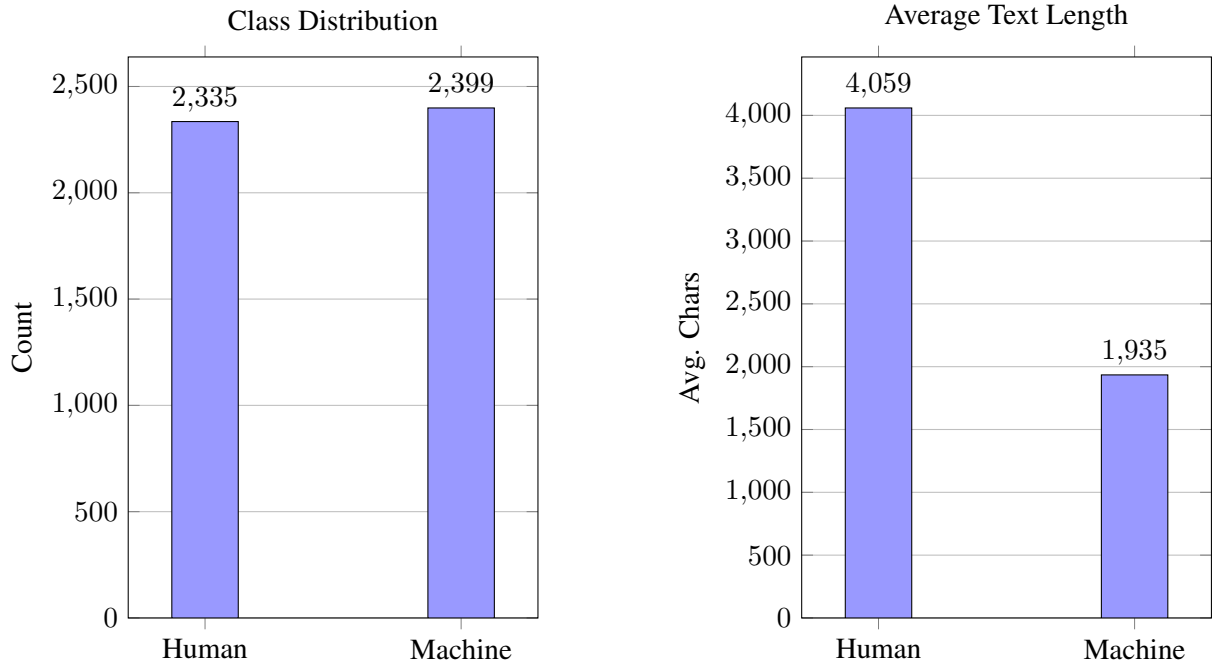


Figure 1: Statistics of the AraGenEval training dataset. The classes are well-balanced, but human-written texts are more than twice as long as machine-generated ones.

Model	F1-Score	Accuracy	Precision	Recall	Specificity	Balanced Acc.
XLM-RoBERTa	0.7701	0.760	0.7390	0.804	0.716	0.760
CAMeLBERT	0.7290	0.710	0.6842	0.780	0.640	0.710
AraELECTRA	0.6180	0.550	0.5369	0.728	0.372	0.550

Table 1: Official results on the AraGenEval test set. XLM-RoBERTa achieved the best performance across all metrics.

All models were then used to generate predictions for the official ‘test_unlabeled.csv’ file.

5.2 Hyperparameters

Models were fine-tuned using the Hugging Face ‘transformers’ library (Wolf et al., 2020). Key hyperparameters are detailed in Table 2.

Hyperparameter	Value
Learning Rate	2e-5
Batch Size (per device)	4
Optimizer	AdamW
Weight Decay	0.01
Max Sequence Length	512
Epochs (AraELECTRA)	4
Epochs (CAMeLBERT)	4
Epochs (XLM-RoBERTa)	5

Table 2: Key hyperparameters for fine-tuning.

5.3 Evaluation Metrics

The primary metric was the macro F1-score. We also report accuracy, precision, recall, specificity,

and balanced accuracy as provided by the official evaluation script.

6 Results

6.1 Quantitative Findings

Our systems yielded varied performance on the official test set, with XLM-RoBERTa emerging as the strongest model. The final results are summarized in Table 1, which led to our 5th place finish.

6.2 Analysis

The most significant finding is that the multilingual XLM-RoBERTa model outperformed both specialized Arabic models. This suggests that the broader and more diverse pretraining corpus of XLM-R may have equipped it with more generalizable features for distinguishing the subtle artifacts of machine generation. As our data analysis showed, the human and machine classes have distinct lexical profiles; XLM-R’s exposure to a vast range of topics and styles in 100 languages likely made it

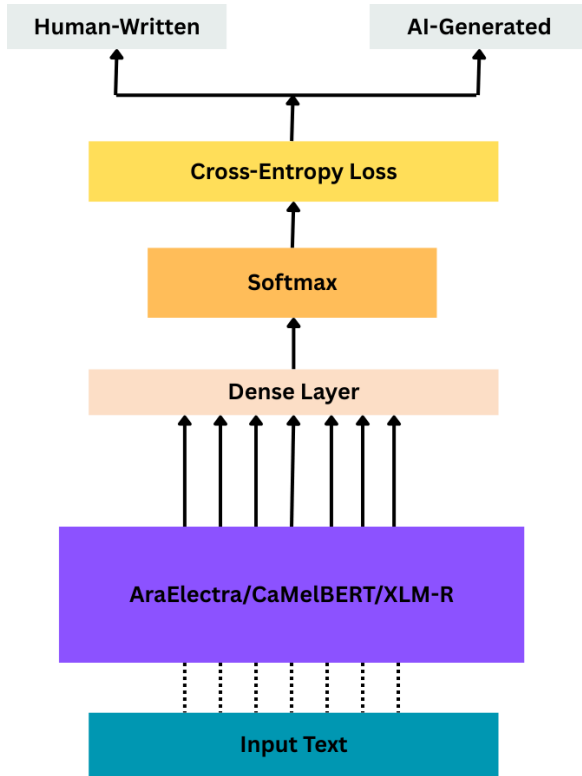


Figure 2: Overview of our comparative system. Input text is processed in parallel by three separate fine-tuned models. AraELECTRA’s pipeline includes an additional text normalization step.

more adept at capturing these stylistic and topical differences.

In contrast, AraELECTRA performance was notably lower. We hypothesize that our aggressive text normalization and diacritic removal, intended to simplify the task, was detrimental. By stripping these features, we likely removed fine-grained signals (e.g., stylistic choices in vocabulary, specific named entities) that our data analysis identified as crucial differentiators between the news-focused human texts and the more formal machine texts. CAMeLBERTE provided a strong baseline but could not match the generalization of XLM-R.

6.3 Error Analysis

While a detailed error analysis was not conducted, the performance gap suggests clear avenues for investigation. The lower precision of all models compared to their recall indicates a tendency to misclassify human text as machine-generated. We hypothesize that errors may stem from domain mismatch or from human-written text that is formulaic or stylistically simple, thus resembling patterns typical of AI generation. Future work should focus on

a qualitative analysis of these false positives.

7 Conclusion

In this paper, we presented our comparative approach for the AraGenEval Shared Task, which resulted in a 5th place ranking. Our experiments showed that the multilingual XLM-RoBERTa model is surprisingly effective for Arabic AI-generated text detection, outperforming specialized monolingual models. Our data analysis revealed significant differences in text length and lexical choice between classes, which likely played a key role in model performance.

Our primary limitation was the suboptimal performance of the AraELECTRA model, likely due to a counterproductive preprocessing strategy. Future work should explore less aggressive text normalization, experiment with model ensembling, and perform a detailed error analysis to better understand the failure modes on this nuanced task.

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